



• PLC PROGRAMMING • LESSON 01

Introduction to PLC Programming

Automate Machines & Industrial Processes

Lesson 1 of 30 | PLC Programming Program | Beginner

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- What is a PLC and why every factory depends on one
- PLC scan cycle — how a controller executes logic in real time
- Hardware architecture — CPU, I/O modules, power supply, rack
- IEC 61131-3 programming languages — Ladder, FBD, ST, SFC
- Industrial applications and PLC career pathways

01 | What Is a PLC?

A **Programmable Logic Controller (PLC)** is a ruggedised industrial computer designed to control machines and processes in real time. First introduced by Dick Morley in 1969 to replace relay panels in automotive assembly lines, PLCs are now the backbone of industrial automation — found in factories, power plants, water treatment facilities, oil refineries, and building management systems worldwide.

Unlike a regular PC, a PLC is built for **harsh environments**: it tolerates extreme temperatures, vibration, humidity, and electrical noise. It executes its control program in a deterministic cycle measured in milliseconds — making it reliable enough to control safety-critical systems like emergency shutdowns and fire suppression.

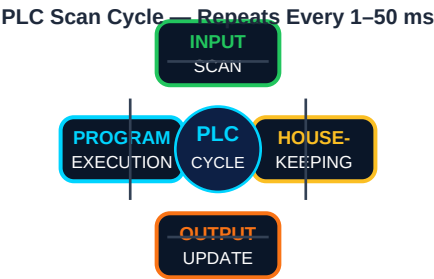
WHY PLCs MATTER — The Scale of Industrial Automation

There are more than 70 million PLCs installed worldwide. Every car you drive was assembled by machines controlled by PLCs. Every bottle of water you drink was filled and capped by PLC-controlled conveyors. Every elevator you take is managed by PLC logic. PLCs represent the bridge between physical machines and the digital control systems that manage them.

02 | The PLC Scan Cycle — How a PLC Executes Logic

A PLC does not run its program once — it runs it continuously in a repeating loop called the **scan cycle**. Each cycle, the PLC reads all inputs, executes the control program logic, and updates all outputs. This cycle repeats every 1–50 milliseconds, giving the PLC near-real-time response to changing field conditions.

The PLC Scan Cycle — 4 Phases:

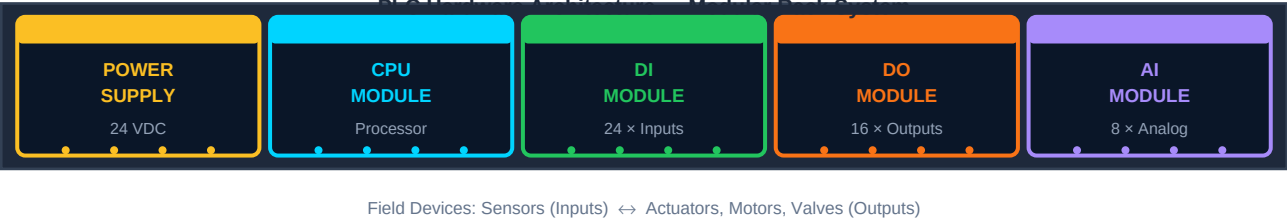


Phase	What Happens	Duration	Key Point
1. Input Scan	PLC reads the state of all digital and analog input terminals and stores values in memory	< 1 ms	Inputs are frozen for the entire program scan — no changes mid-cycle
2. Program Execution	CPU executes user program logic top-to-bottom, rung by rung, using stored input values	1–10 ms	Longer programs = longer scan time; keep programs efficient
3. Output Update	CPU writes calculated output values from memory to physical output terminals	< 1 ms	Physical devices (motors, solenoids) only change state here
4. Housekeeping	PLC checks its own health: watchdog timer, comms, diagnostics, and memory checks	< 1 ms	Watchdog resets; if it misses, PLC faults and stops safely

Control Method	Flexibility	Reprogramm ing	Speed	Reliability	Best For
Relay Panel	Very low	Rewire hardware	Electromechani cal	Moderate — wear parts	Legacy simple circuits
PLC	High	Software change	1–50 ms cycle	Very high — designed for industry	Industrial automation
PC / IPC	Very high	Software change	Variable	Moderate — OS overhead	Data-heavy, SCADA, HMI apps
Microcontroller	High	Reprogram chip	Microseconds	High — embedded	Small embedded products

03 | PLC Hardware Architecture

A PLC system is built from modular hardware components installed in a **rack (chassis)**. Each module slots into the backplane, which provides power and communication between modules. This modular design allows engineers to customise the PLC for each application — from a simple 8-input machine to a complex distributed control system with hundreds of I/O points.



Module	Function	Key Specification
Power Supply (PS)	Converts AC mains (120/240V) to DC (5V/24V) for rack and modules	Current rating (A), redundancy option
CPU Module	Executes the control program, manages memory, handles comms and scan timing	Scan time (ms), memory size, comms ports (Ethernet, Profibus, Modbus)
Digital Input (DI)	Reads ON/OFF states from switches, sensors, pushbuttons, limit switches	Number of channels, voltage rating (5V/24V/110V/240V), input type (sourcing/sinking)
Digital Output (DO)	Switches devices ON/OFF: motors (via starter), solenoids, lamps, alarms	Output type (relay/transistor/triac), current rating, channel count
Analog Input (AI)	Reads variable signals from sensors: temperature (4–20 mA), pressure, flow	Resolution (12/16 bit), signal range (0–10V or 4–20 mA), channel count
Analog Output (AO)	Sends variable control signals to drives, control valves, damper actuators	Resolution, output range, update time

04 | IEC 61131-3 Programming Languages

The IEC 61131-3 standard defines five official PLC programming languages. All major PLCs (Siemens, Allen-Bradley, Mitsubishi, Omron) support these languages. Most industrial programs are written in Ladder

Diagram (LD) or Structured Text (ST), but complex processes often combine multiple languages in one project.

Language	Abbreviation	Visual Style	Best Used For	Typical Users
Ladder Diagram	LD	Horizontal rungs — like relay circuits	On/Off control, interlocks, motor starters	Electricians, maintenance, traditional PLC engineers
Function Block Diagram	FBD	Interconnected function blocks (boxes)	Process control, PID loops, data routing	Process & instrumentation engineers
Structured Text	ST	High-level text code (like Pascal)	Math, data processing, complex algorithms	Software / computer science backgrounds
Sequential Function Chart	SFC	Flowchart-style steps and transitions	Sequential machine processes, batch control	Process engineers, batch systems
Instruction List	IL	Assembly-like text mnemonics	Legacy systems, low-level optimization	Legacy maintenance — largely deprecated

05 | Ladder Logic — The Language of Industrial Automation

Ladder Diagram (LD) is the most widely used PLC language in industry. Its visual layout was deliberately designed to look like the relay control circuits it replaced, making it intuitive for electricians. Each horizontal **rung** represents one logic equation: conditions (contacts) on the left determine whether the output (coil) on the right is energised.

Start/Stop Motor Control — Classic Ladder Rung:



READING LADDER LOGIC

Think of each rung as asking a question: "IF (I0.0 Start is pressed) AND (I0.1 Stop is NOT pressed) THEN turn ON (Q0.0 Motor)." Contacts read left-to-right like AND logic. Branches stacked vertically are OR logic. The output coil activates only when a continuous electrical path exists from left rail to right rail through all the conditions on that rung.

06 | PLC Applications Across Industries

Industry	Typical PLC Applications	Common PLCs Used
Manufacturing	Conveyor control, robot coordination, press control, assembly sequencing	Siemens S7, Allen-Bradley ControlLogix, Mitsubishi Q-Series
Oil & Gas	Pump control, pipeline monitoring, emergency shutdown (ESD), flare systems	Allen-Bradley, Siemens S7-400, Honeywell

Water & Wastewater	Pump sequencing, chemical dosing, filtration control, tank level control	Schneider Modicon, Siemens S7-300, Omron
Building Automation	HVAC control, elevator logic, fire alarm systems, access control	Siemens, Schneider, Johnson Controls
Food & Beverage	Filling machines, packaging lines, pasteurisation, CIP (Clean-In-Place)	Allen-Bradley, Omron, Mitsubishi
Power Generation	Turbine control, generator synchronisation, substation automation	ABB, GE, Siemens SIMATIC

07 | PLC Career Pathways

Role	Responsibilities	Key Skills	Avg Salary (USD/yr)
PLC Programmer	Write, test, and debug PLC control logic for new and existing systems	Ladder Logic, FBD, ST, TIA Portal / Studio 5000	\$55,000 – \$90,000
Automation Engineer	Design full automation systems: PLC + HMI + SCADA + drives	System integration, process knowledge, networking	\$70,000 – \$110,000
Controls Engineer	Specify and commission control hardware and software for projects	Control panels, I/O design, motor drives, instrumentation	\$65,000 – \$105,000
SCADA / HMI Developer	Build operator interfaces and supervisory monitoring systems	Ignition, WinCC, FactoryTalk, SQL, Python	\$65,000 – \$100,000
Maintenance Technician	Troubleshoot faults in PLC-controlled systems, modify programs on-site	Fault diagnosis, ladder reading, multimeter, drives	\$45,000 – \$75,000

Reflection:

"Walk through any industrial facility, factory, or building near you. Identify three machines or processes that are controlled automatically. For each one, consider: what inputs (sensors) does it read, what outputs (actuators) does it control, and what logic must it follow? You have just described three PLC programs."

Action checklist:

Next up — PLC-002:

PLC Hardware Deep Dive — CPU architecture, I/O addressing, wiring inputs and outputs to field devices.

Community & support:

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